

Adil Faizi

adil.fzi@outlook.com | linkedin.com/in/adilfaizi/ | github.com/AdilFaizi | (647) 335-5251

Education

York University

Bachelor of Engineering, Electrical Engineering (Honours)

Sept 2023 - May 2027

- Course Work: Power Systems, Linear Control Systems, Applied Electromagnetism, Signal And Systems, Electric Circuit and Devices
- Co-op Eligibility: 4/8/12/16-month terms

Skills

Programming: MATLAB, Python, C, Java, Bash, Arduino

CAD & Circuit Design: AutoCAD, Revit, KiCad, ETAP

Analysis & Simulation: LTspice, MATLAB

Productivity: Microsoft Office (Excel, Word, PowerPoint), Acrobat Reader

Experience

Distribution Planning Engineering Student Hydro One – Toronto, ON

May 2026 – August, 2027

- Developed power system investment scopes for capacity reinforcement, reliability improvements, asset refurbishment, and new load connections, providing technical recommendations supported by detailed documentation.
- Performed power system and protection assessments for customer grid connections, identifying connection alternatives, diagnosing power loss issues, and implementing effective mitigation solutions.
- Collaborated with internal and external stakeholders to resolve technical and process issues while ensuring compliance with power industry standards and utility requirements.

Projects

DSP Filter Design and Audio Restoration Project | MATLAB

- Designed a bandpass FIR filter in MATLAB meeting strict passband and stopband ripple specifications, computed filter length, coefficients, and frequency response.
- Analyzed corrupted audio using frequency-domain techniques to identify and isolate noise components.
- Implemented time- and frequency-domain noise removal methods and verified performance via ripple/attenuation metrics and listening tests.

Ultrasonic Distance Checker with I2C LCD Display | Arduino

- Designed a sensor-based instrumentation system using ultrasonic measurement, signal timing, and control logic to acquire and validate real-time distance data.
- Programmed an Arduino Uno R4 Minima to acquire, process, and update distance readings at fixed intervals.
- Implemented efficient I2C communication to ensure smooth, responsive LCD updates.

Remote Controlled Car | Embedded C and Python

- Implemented an RC car system using microcontrollers and programming in Embedded C and Python, enabling wireless control and autonomous features
- Developed an object-oriented control framework to manage propulsion, steering, and sensor integration
- Integrated real-time messaging to log diagnostics such as speed, battery level, and location to a .txt file

Automated Irrigation System | MATLAB and Java

- Built a Java–MATLAB system that automates irrigation by detecting soil conditions and simulating large scale water-distribution logic.
- Integrated CO₂ and sunlight sensors, plus real-time mobile alerts via a custom messaging interface

- Optimized system performance through iterative testing to improve sensing accuracy and automated decision-making.